

Technical Specification: IntelliGuard Detection Mathematics

This document outlines the mathematical framework governing the IntelliGuard multi-layered prompt injection detection pipeline. The system utilizes a hierarchical ensemble of transformer architectures to map natural language inputs into high-dimensional vector spaces for classification.

1. Mathematical Foundations of the SPINE Layer

The SPINE layer utilizes a DistilBERT architecture, which is a distilled version of the BERT (Bidirectional Encoder Representations from Transformers) model. Mathematically, this layer focuses on rapid pattern recognition by approximating the full transformer distribution.

1.1. The Self-Attention Mechanism

The core of the detection logic resides in the scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Where:

- **Q (Query)**: Represents the current token.
- **K (Key)**: Represents all tokens in the input sequence.
- **V (Value)**: Contains the semantic information to be aggregated.
- d_k : The dimensionality of the keys, used as a scaling factor to prevent gradient vanishing.

1.2. Knowledge Distillation Loss

DistilBERT is trained to minimize a triple loss function, ensuring the "SPINE" retains the performance of larger models while reducing latency:

$$L_{\text{total}} = \alpha L_{\text{ce}} + \beta L_{\text{distill}} + \gamma L_{\text{cos}}$$

This allows the model to produce a safety probability $P(S)$ with significantly lower time complexity.

2. Semantic Vector Space of the BRAIN Layer

The BRAIN layer employs XLM-RoBERTa, which operates in a cross-lingual embedding space. This is critical for detecting obfuscated or multilingual injections (e.g., Base64, Hex, or translated

attacks).

2.1. High-Dimensional Embeddings

Input tokens are mapped to a vector $\mathbf{h} \in \mathbb{R}^{768}$. The detection of an injection is treated as a distance calculation in the latent space. If a vector $\mathbf{h}_{\text{input}}$ falls within the hypersphere defined by $\mathbf{h}_{\text{malicious}}$, the system flags the intent.

Component	Mathematical Role	Impact on Score
Softmax Output	Converts raw logits to $[0, 1]$ probability.	Determines individual layer confidence.
Cross-Entropy Loss	Measures divergence between predicted and actual label.	Optimizes detection accuracy during training.
Cosine Similarity	Measures the angle between two embedding vectors.	Identifies semantically similar attack vectors.

3. The Ensemble Aggregator

The final IntelliGuard verdict is not a single model output, but a weighted ensemble. This mitigates False Positives from a single layer.

3.1. Final Weighted Score Formula

The final threat score T_{final} is calculated as:

$$T_{\text{final}} = (w_1 \cdot P(\text{SPINE}) + w_2 \cdot P(\text{BRAIN})) / (w_1 + w_2)$$

Where:

- w_1 (**SPINE weight**): Typically optimized for low-latency pattern recognition.
- w_2 (**BRAIN weight**): Higher weight is often assigned to this layer due to its superior semantic understanding of complex injections.

4. Hardware Acceleration Math (MI300X Context)

When deployed on high-density accelerators like the AMD MI300X, the system leverages massive parallelization for these matrix operations. The throughput R is defined by:

$$R = (N_batches \cdot N_tokens) / Latency_ms$$

The high memory bandwidth allows the model to process the 500-sample batch scans at near-peak theoretical flops, ensuring that the 4-layer detection pipeline does not bottleneck the overall RAG system.